Toy-car investigation: digital graphing and analysis

STUDENT INFORMATION AND EXAMPLES

From measured data to a position–time model

Use the five measured elapsed-time readings at each position to calculate one mean time. Keep the complete raw table. Enter the pairs (mean time, position) into Desmos: horizontal = time in s; vertical = position in m.

1. Open https://www.desmos.com/calculator. Select + → Table, and enter mean times in x_1 and positions in y_1.
2. Use the graph settings (wrench) to select a suitable view. Add text or a screenshot with the physical axis names and units when submitting.
3. Show a suitable smooth best-fit curve. You may test a quadratic model, $y_1 \sim a\,x_1^2 + b\,x_1 + c$, if this fits the measured trend; it is a model, not a guarantee.
4. Estimate the initial and final gradients using tangents to the fitted trend or narrow-region secants. State times and units.

Building a velocity–time graph

For adjacent data points, compute interval average velocity = $\Delta x/\Delta t$. Plot it at midpoint time = $(t_{start} + t_{end})/2$. Repeat for each interval.

These are interval-average velocities; they estimate instantaneous velocity near each midpoint when intervals are short and motion changes smoothly.

Plot velocity (m/s) vertically against midpoint time (s) horizontally. If velocity increases roughly linearly, the v–t gradient (m/s^2) estimates acceleration.

“Velocity change per second” means $\Delta v/\Delta t$, the acceleration. “Velocity change per cm” means $\Delta v/\Delta x$ over a specified stretch; it need not be constant even if acceleration is.

Reflection and evaluation

Describe the evidence for changing velocity. Refer to features of your x–t and v–t graphs and actual data. Distinguish what the data show from possible explanations.

Discuss one uncertainty or method limitation you noticed and propose a specific practical improvement with a reason.

Discuss one quantity that may not have been fully controlled during the investigation, and explain how it could have affected the results.

Desmos reference

Graphing calculator: https://www.desmos.com/calculator

Table instructions: https://help.desmos.com/hc/en-us/articles/4405489674381-Tables